

Directions: For each of the studies below, answer the questions that follow. Then, in the graphic organizer, made a check in the correct scope of the study.

Study #1: Lack of sleep is associated with increased risk of cardiovascular disease, depression, and other health concerns. A new study now shows that lack of sleep also affects social interactions, making people less willing to help others (empathy). Scientists placed 24 healthy volunteers in a functional magnetic resonance imager (fMRI) to scan their brains after eight hours of sleep and then again after a night of no sleep. The order that they received the treatment of sleep vs no sleep was randomly selected with the flip of a coin. They found that areas of the brain that form the theory of mind network, which is engaged when people empathize with others or try to understand other people's wants and needs, were less active after a sleepless night. "Here, we found that a decrease in the quality of someone's sleep from one night to the next predicted a significant decrease in the desire to help other people from one subsequent day to the next." Source: <https://www.sciencedaily.com/releases/2022/08/220823143827.htm>

Is this an observational study or an experiment? Explain.	Which box of inference does this study fall under? <table border="1"><tr><td></td><th>Inference about cause and effect</th><th>No Inference about cause and effect</th></tr><tr><th>Inference about the Population</th><td></td><td></td></tr><tr><th>No Inference about the Population</th><td></td><td></td></tr></table>		Inference about cause and effect	No Inference about cause and effect	Inference about the Population			No Inference about the Population		
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Can this study make inferences about cause and effect or just an association between variables? Explain.										

Study #2: A study of over 26,000 middle-aged women reveals those with a vegetarian diet had a 33% higher risk of hip fracture compared to regular meat-eaters. The women, ages 35–69 years, and randomly selected from a large pool of UK middle-aged women, were classified as regular meat-eaters (≥ 5 servings/week), occasional meat-eaters (< 5 servings/week), pescatarian (ate fish but not meat), or vegetarian (ate neither meat nor fish) based on questionnaire they completed at the beginning of the study. Twenty years later, their medical records were examined to determine if they suffered a hip fracture during the course of the study. Source: <https://bmcmmedicine.biomedcentral.com/articles/10.1186/s12916-022-02468-0>

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Study #3: Dogs and humans clearly have a special bond. But do dogs, like humans, produce more tears at times when they are flooded with emotion? A new study says that, indeed, the eyes of our canine companions do well up with tears. A random sample of dog owners was selected from the US population and the Schirmer tear test (STT) was performed. The participants were randomly placed in to one of two groups: reunited with their owners and reunited with a non-owner. They measured tear volume in dogs before and after reunions with owners and non-owners. Tear volume increased significantly during reunion with the owner, but not with a non-owner. Scientists say that the findings came as a surprise. "We had never heard of the discovery that animals shed tears in joyful situations, such as reuniting with their owners, and we were all excited that this would be a world first!" Source: <https://www.sciencedaily.com/releases/2022/08/220822130249.htm>

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Study #4: Teens and adults in the United States differ on a key issue tied to online speech and its consequences. Two separate samples of US teenagers (aged 13 to 17 years old) and US adults (18 and older), showed that 62% of teenagers say people being able to feel welcome and safe online is more important than people being able to speak their minds freely online, while 57% of adults hold the same view. When looking at party lines, the differences between the two age groups become even larger. 71% of democratic leaning teens believe more in feeling safe online, while only 49% of republican leaning teens believe the same view. For adults, 62% of democratic leaning adults believe in feeling safe online is more important than free speech, while 36% of republican leaning adults feel the same. Source: <https://www.pewresearch.org/fact-tank/2022/08/30/more-so-than-adults-u-s-teens-value-people-feeling-safe-online-over-being-able-to-speak-freely/>

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